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Increasing maize yield and farming benefit through integrated plant management in Madura Island East Java

W Handayati, E Latifah, Sugiono, H Subagio, D Sihombing and R U Fitria
Assessment Institute of Agriculture Technology East Java, Jl. Raya Karangploso KM 4 Malang

e-mail: wahyuhandayati@yahoo.com

Abstract. Maize is one of the primary plant cultivations on Madura Island and is usually cultivated by farmers with traditional technology and using open-pollinated local varieties. According to this condition, an experiment and a demo plot for local farmers were conducted in Larangan Slampar village Tlanakan District Pamekasan Regency. The experiment was arranged in randomized block design. The treatment was integrated plant management (IPM) using HJ21 and Bima14 hybrid and as control was conventional technology applying local Madura variety. The analyzed data indicated that IPM technology could increase plant yield and farming benefits. The highest harvest yield was shown by IPM1 with Bima 14 variety up to 6.35 ton ha-1, meanwhile existing technology (local Madura variety) only 1.22 ton ha-1 or the productivity was increased up to 422,40 %. The results of the farming analysis showed that benefit with IPM1 reached Rp 15,520,500 and B/C ratio 1,35; while existing technology only Rp 2,168,000 and B/C ratio of 0.55.

1. Introduction

Madura Island is one of the maize cultivation centers in Indonesia. The harvest area is quite wide, but the productivity still very low and far below the national production average. According to BPS 2021, the average productivity was only 2,187 tons (Bangkalan), 1,729 tons (Sampang), 2,258 tons (Pamekasan), and 2,278 tons ha-1 (Sumenep) [1], meanwhile the national productivity up to 5,474 tons ha-1 [2]. Supposedly, this condition is affected by conventional technology applied by farmers, especially the local varieties usage. In addition, knowledge of farmers is still lacking about Integrated Plant Management (IPM) and its application in the field. About 88% of farmers in Madura are cultivating the local varieties of maize to fulfill the family's food supplies. Some farmers reason to use local varieties, namely seed availability, lower farm costs, longer shelf life, lower post-harvest cost, and higher selling prices [3][5].

Integrated maize crop management aimed to increase maize productivity, maintain sustainability, and improve production efficiency. This development must consider the local resource conditions [6] [7]. In this regard, the technological components that have been obtained from various studies are assembled into one technology package that can provide a synergistic effect. And the application must be made with an integrated maize management system. Implementation of the components of production technology such as superior varieties, quality seeds, optimal plant populations, and efficient fertilization, must be adjusted to the land conditions and socio-economic farmer communities [8]. On marginal lands, such as on Madura Island, maize productivity can be significantly increased with the support of IPM technology innovation.

To get maximum productivity, then the superior varieties must be combined with others planting techniques. Because the highest yields can be resulted by plants if the superior varieties get optimal accessibility to the water, fertilizer, and other planting techniques combination. So far, the cultivation technology components of maize have been used partly by the farmer, especially in the low productivity field. Therefore, there is no significant effect on farming production. A previous study on IPM technology application with new hybrid varieties showed the increasing yield of up to 8.8 tons on Bima 4, 8.3 ton Bima 5, and 8.4 tons on Bisi 2, meanwhile the existing only 4.90 – 5.80 tons ha^{-1} [9].

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To accelerate the IPM technology adoption by farmers, it is necessary to introduce this technology through field activities involving farmers and farmers' groups as the main actors, and researchers or extension workers as facilitators and supervisors in the application of technology. These activities must be designed in such a way that farmers are interested in replicating them in subsequent farming activities. In this regard, the research activities have been carried out as well as demonstration plots for farmers. The study was aimed to validate the increase of maize productivity, farming benefits, or feasibility by the determination of cultivation technology with an integrated plant management approach.

2. Materal and Method

The study was done in the dryland of Larangan Slampar village Tlanakan District Pamekasan Regency. This assessment was arranged in a randomized block design with six replications, The treatments were the cultivation technology applied i.e., integrated plant management (IPM) approach using HJ 21 and Bima 14 varieties, and existing technology using local variety (Madura) as control. Details of each treatment in the field were assembled in Table 1.

Table 1. Treatment arrangement of IPM technoloy

Activity	Treatments		
	IPM1	IPM2	Existing (control)
1. Land cleaning	yes	yes	yes
2. Land processing	yes	yes	No
3. Varieties	Bima 14	HJ 21	Local variety (cv Madura)
4. Seed (kg/ha)	20	20	15
5. Plant space (cm)	70 x 20	70 x 20	40 x 40
6. Organic manure (ton/ha)	1	1	0
7. An-organic manure (kg/ha)	Urea 300 kg; NPK Ponska 200 kg	Urea 300 kg; NPK Ponska 200 kg	Urea 100 kg
8. Irigation	6 x	6 x	0
9. Weed control	Herbicide 1 application in 2 week after palnting	Herbicide 1 application in 2 weeks after palnting	No
10. Pest and disease control	Pesticide 1 application	Pesticide 1 application	No

The land has tilled by hand tractor and flattened with a hoe. The plant hole was made with a wood digger. After seed planting, the hole was covered by organic fertilizer. Inorganic fertilizer dose was based on the Dry Soil Test Kit (DSTK) and Leaf Color Chart (LCC) analyses. Application of fertilizer was carried out twice, as much as 1/3 dose at 14 days after planting, and the remaining at 30 days after planting. Plant maintenance was done according to the treatment arrangement. The observed variables consist of plant growth, yield components, yield, and analyses of farming.

Analyzed of variance was done for tabulated data and continued with the LSD test at a level of 0.05 if between the treatments had a significant difference. All agronomic variables observed was The farmer income or financial benefit by IPM technology treatment was estimate through a simple farming analysis.

To measure the feasibility of improvement technology (IPM) used MBCR (Margin Benefit Cost Ratio) analysis. If the value of MBCR > 1 then the improvement technology (IPM) must be replacing the technology applied by farmers. It can be simply derived with a formula:

$$MBCR = \frac{Total\ Benefit\ of\ IPM - Total\ Benefit\ of\ exising}{Total\ cost\ of\ IPM - Total\ cost\ of\ existing}$$

3. Result and discussion

3.1. The field agroecoclimate conditions

The assessment was placed in the dry lowland with dry climate conditions. The land was a clay texture and was classified as alluvial soil. DSTK analyses indicated that the land contained a low P, K, and C-organic content and neutral pH. A different study reported that the soil in this district was very poor in nutrients element i.e., N 0.01%, P205 1.29 mg kg-1, K 0.16 m/100 g, C-organic 0.46, and pH 6.2 [10].

Generally, the rainy period in this location was only from November to April. In the assessment, there was no rain from the beginning of planting until harvesting, Therefore, the water necessity was fulfilled only from watering. Daily temperatures range from 28-32°C and humidity 65-70%.

3.2. Analysis of variance for parameters observed

Anova for all parameters was presented in Table 1. There was no significant effect (P > .05) by the blocking data to all parameters observed. Then, the blocking effect was ignored, so that the degree of freedom Error (df) can be increased. All agronomic characters were significantly influenced by the treatments, except for the weight of net kernels (Table 2).

Table 2. Anova for maize agronomic characters

Parameter	Source of Variation	Degree of freedom (Df)	Sum of squares	Means squares	F	Sig.
Plant height (cm)	Between groups	2	25,111.80	12555.90	80.08	7.01E-07
	Within groups	15	1,808.73	204.95		
	Total	17	26,920.53	1583.56		
Stalk diameter (cm)	Between groups	2	10.31	5.16	51.29	5.53E-06
	Within groups	15	1.87	0.27		
	Total	17	12.18	0.72		
Leaf length (cm)	Between groups	2	8066.09	4033.05	155.72	2.91E-08
	Within groups	15	367.03	47.51		
	Total	17	8433.12	496.07		
Leaf width (cm)	Between groups	2	110.22	55.11	102.80	2.15E-07
	Within groups	15	11.92	1.85		
	Total	17	122.14	7.18		
Cob length (cm)	Between groups	2	227.94	113.97	26.97	9.36E-05
	Within groups	15	76.59	11.09		
	Total	17	304.53	17.91		
Cob diameter (cm)	Between groups	2	26.71	13.35	87.11	4.71E-07
	Within groups	15	3.35	0.52		
	Total	17	30.06	1.77		
Cob weight (g)	Between groups	2	168409.71	84204.86	156.70	2.83E-08
	Within groups	15	9392.18	1341.06		
	Total	17	177801.89	10458.93		
Kernel weight/cob (g)	Between groups	2	106192.71	53096.36	188.62	1.15E-08
	Within groups	15	3361.560	390.82		
	Total	17	109554.27	6444.37		
Net kernel weight (%)	Between groups	2	0.01	0.00	1.55	0.26
	Within groups	15	0.08	0.01		
	Total	17	0.09	0.01		
100 kernel weights (g)	Between groups	2	2906.34	1453.17	191.52	1.07E-08
	Within groups	15	209.12	34.24		
	Total	17	3115.47	183.26		
Yield (ton)	Between groups	2	82.46	41.23	208.48	7.05E-09
	Within groups	15	3.16	0.43		
	Total	17	85.62	5.04		

3.3. Performance of plant growth

The data showed that plant growth was significantly affected by introduced cultivation technology (IPM). The highest plants were found in IPM1 and IPM2, and it was significantly different from the existing treatment. Stalk diameter was also found in a significant difference between treatments tested. The biggest stalk diameter was found in IPM2 (Table 3).

Observation of leaf showed that there was a significant difference between treatments both on the leaf length and leaf width, especially to existing (Table 3). This finding was similar to the present report that all local Madura varieties had a small plant posture and cob. [11]

Table 3. Effect of IPM technology to plant growth of maize in dry season at Larangan Slampar village Tlanakan district Pamekasan regency.

Treatments (Cultivation technology)	Plant height (cm)	Stalk diameter (cm)	Leaf length (cm)	Leaf width (cm)
IPM1 (Bima 14)	174.83 a	2.81 a	85.53 a	9.59 a
IPM2 (HJ 21)	179.66 a	2.88 a	89.90 a	9.55 a
Existing (local variety)	98.12 b	1.24 b	42.97 b	4.32 b
CV	8.30	13.74	6.99	9.36
SEM	5.11	0.13	2.08	0.30

3.4. Plant yield

The data analyses of the harvest yield component indicated that the cob length was significantly affected by the treatments. The longest cob was found in IPM1 treatment (18.66 cm), and its significant differences from the existing (10.94 cm) (Table 4).

The cob diameter showed that the largest cob was resulted by IPM1, and it has a significant difference to existing. IPM1 also produced the heaviest cobs up to 267.07 g and had a significant difference with the existing (Table 4). This result indicates that varieties had the highest effect on the harvest yield component.

Table 4. Effect of IPM technology to maize yield in dry season at Larangan Slampar Tlanakan Pamekasan

Treatments (Cultivation technology)	Cob length (cm)	Cob diameter (cm)	Cob weight (g)	Kernel weight/ cob (g)	Net kernel weight (%)	100 kernel weights (g)	Yield (ton)	Increased yield of IPM to existing (%)
IPM1 (Bima14)	18.66 a	4.82 a	267.39 a	216.45 a	0.81 a	44.22 a	6.35 a	422.40
IPM2 (HJ 21)	18.30 a	5.13 a	221.81 b	170.69 b	0.78 a	41.64 a	4.70 b	286.38
Existing (local variety)	10.94 b	2.40 b	43.24 c	35.53 c	0.82 a	16.07 b	1.22 c	
CV	12.88	9.51	13.06	11.91	5.13	8.11	10.88	
SEM	0.84	0.16	9.46	6.85	1.69	1.25	0.18	

The heaviest kernel per cob has resulted from IPM1. It was significantly different from IPM2 and existing. The weight per 100 kernels was significantly different from existing (Table 4).

The net weight of the kernel is a comparison of the kernel weight to the cob ear weight before threshed. There was no significant difference between the treatment examined. The heaviest net kernel was obtained by existing (Table 4). This fact indicated that this variety could produce kernels with an optimum weight, even though the maize cobs size of local Madura was a lot smaller than other varieties.

The yield data in Table 4 showed that IPM1 produces the highest yielding 6.35 tons ha-1, and it was a significant difference to IPM2 and a very significant difference to existing which just produced 1.22 tons ha-1. Furtherly, this treatment could enhance the yield by 422.40% than existing technology. Another study also reported the same result that yields on IPM technology (cooperative farmers) were higher than conventional technology (non-cooperative farmers). Srikandi Kuning and Sukmaraga varieties gave yields of 7.45 and 7.90 tons ha-1 respectively or 119.28% higher than farmers' technology with the local variety

Manado Kuning (only 3.50 t ha-1) [12]. The character expression will be optimum if the crop was planted in a suitable environment, as in the IPM treatment condition. Inversely, if an environment of growth was inappropriate, the character performance will not be optimum. A character performance with low heritability has a significant environmental effect. Hence, its appearance will be easy to change. However, the high heritability character has a little environmental effect. Therefore, the performance will be quite stable [13].

Table 5. The correlation coefficients of harvested yield and agronomic parameters

Parameters	Plant height (cm)	Stalk diameter (cm)	Leaf length (cm)	Leaf width (cm)	Cob length (cm)	Cob diameter (cm)	Cob weight (g)	Seed weight/ cob (g)	Net kernel weight (%)	100 kernel weight (g)	Sampling yield (kg)/5,46 m2
Plant height (cm)	1										
Stalk diameter (cm)	0.916	1									
Leaf length (cm)	0.911	0.889	1								
Leaf width (cm)	0.903	0.803	0.921	1							
Cob length (cm)	0.860	0.864	0.840	0.763	1						
Cob diameter (cm)	0.918	0.869	0.904	0.825	0.816	1					
Cob weight (g)	0.920	0.897	0.934	0.865	0.860	0.885	1				
Seed weight/ cob (g)	0.897	0.864	0.919	0.909	0.815	0.852	0.977	1			
Net kernel weight	-0.207	-0.275	-0.261	0.041	-0.359	-0.266	-0.277	-0.073	1		
100 kernel weight (g)	0.928	0.931	0.953	0.856	0.922	0.912	0.982	0.947	-0.340	1	
Sampling yield (kg/5,46 m2)	0.909	0.912	0.921	0.902	0.861	0.897	0.963	0.977	-0.108	0.958	1
Harvest yield (ton/ha)	0.890	0.865	0.878	0.893	0.855	0.816	0.932	0.959	-0.034	0.918	0.959

Correlation analyses among agronomic variables indicated that all the parameters were correlated to the yield, except the weight of the net kernel). The plant height, the cob length, diameter, and weight; and kernel weight/cob, and 100 kernel weights had the highest coefficients correlation to the harvest yield (Table 5. The same result in earlier research showed that harvest yield was positively correlated with plant height, ear height, and field weight and with high genotypic and phenotypic correlation coefficients [14]. A different report quoted that kernel was positively correlated and had a significant correlation with plant and flag leaf height, 300-kernel weight, total kernel per row, and per ear [15].

3.5. Farming Analysis

The results of the farming analysis showed that the IPM1 assembly technology provided the highest income or revenue up to Rp 26,987,500, reaching the R/C ratio of 2.35 while in existing only 6,100,000 with an R/C ratio of 1.55. Although the production costs used in IPM1 increase, it can provide higher profits than existing. It looks at the B/C ratio of 1.5 in IPM1 and 0.55 in existing technology with a benefit difference reaching Rp 13,352,000 for IPM1.

Based on feasibility analysis, the highest MCBR value showed by technology introduced IPM1 (Bima 14 variety) reached 1.77. Although the costs incurred were higher than existing technology, the IPM1 technology assembly is suitable for development because it can provide higher profits and incomes to farmers. Thus, the application of IPM1 with Bima 14 can be recommended as a location-specific technology.

This result is in line with previous research that the gaining of income or profits in IPM technology (Rp 8,687,500) increased 197 % than existing technology (Rp4,390,000) [12]. Another study showed that IPM with different varieties also affects the cost of production and maize yields. Farming cost about Rp 6,013,140 for BISI and Rp 3,826,620 for Lamuru varieties with productivity of 4,785 kg and 4,607 kg ha-1 respectively [16]. The study in Sumenep showed that maize yields 4.93 tons with a profit up to Rp 6,944,099 [11]. Another study showed that 3.90 tons ha-1 of corn yields resulted in a profit of Rp 2,104,950 [17].

Table 6. Farming analisys of IPM technology application on maize cultivation in dry season at Larangan Slampar village Tlanakan district Pamekasan regency

Description	Units	IPM1 (Bima 14)			IPM2 (HJ 21)			Existing (lokal Madura)		
		Volume	Unit price (Rp)	Total (Rp)	Volume	Unit price (Rp)	Total (Rp)	Volume	Unit price (Rp)	Total (Rp)
Cost										
1. Seed	kg	22	56,000	1,232,000	22	56,000	1,232,000	-	-	-
2. Organic fertilizer	kg	1,000	500	500,000	1,000	500	500,000		500	-
3. NPK Ponska fertiizer	kg	200	2,300	460,000	200	2,300	460,000	100	2,300	230,000
4. Urea fertilizer	kg	300	1,800	540,000	300	1,800	540,000	100	1,800	180,000
5. Pesticide	package	1	500,000	500,000	1	500,000	500,000	-	500,000	-
6. Land processing (hand tractor)	package	1	1,000,000	1,000,000	1	1,000,000	1,000,000	1	1,000,000	1,000,000
7. Planting	worker	20	40,000	800,000	20	40,000	800,000	20	40,000	800,000
8. Plant maintanance (fertilizer application, weeding, pest/disease	worker	50	40,000	2,000,000	50	40,000	2,000,000	20	40,000	800,000
9. Irrigation	package	6	500,000	3,000,000	6	500,000	3,000,000		500,000	-
8. Harvesting	worker	20	40,000	800,000	20	40,000	800,000	20	40,000	800,000
9. Thresher	kg	6,350	100	635,000	4,700	100	470,000	1,220	100	122,000
A. Total cost	Rp			**11,467,000**			**11,302,000**			**3,932,000**
B. Revenue	kg	6,350	4,250	**26,987,500**	4,700	4,250	**19,975,000**	1,220	5,000	**6,100,000**
C. Benefit	Rp			**15,520,500**			**8,673,000**			**2,168,000**
D. Δ Benefit of IPM to existing technology	Rp			**13,352,500**			**6,505,000**			
E. R/C ratio				**2.35**			**1.77**			**1.55**
F. B/C ratio				**1.35**			**0.77**			**0.55**
G. MCBR				**1.77**			**0.88**			

4. Conclusions

Application of IPM technology, especially IPM1 with Bima 14 variety could increase the harvesting yield and farming benefit or farmer income. The harvesting yield obtained up to 6,35 ton ha-1, and it increased up to 422.40% compared to existing. The revenue or income increased up to Rp 26,987,500 with an R/C ratio of 2,35. The farming benefits about Rp 15,520,500 with a B/C ratio of 1,35. This benefit increases up to Rp 13,352,500 than existing. The highest MCBR value in technology introduced IMP1 (Bima 14 variety) reached 1.77.

This IPM technology assembly using Bima 14 and others superior hybrid varieties was recomendated to replace the conventional tehnology

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